HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented hydraulic models for flood risk assessments across **Ireland and the UK**, integrating complex terrain data and rainfall-runoff simulations.
- Managed end-to-end delivery of **hydraulic modelling projects**, from initial client scoping and proposal development through to detailed design reports and stakeholder presentations.
- Collaborated with **multidisciplinary teams** including environmental scientists and engineers to deliver integrated sustainability solutions for infrastructure development.
- Developed and executed **TUFLOW models**, including advanced **TUFLOW FV** scenarios, to simulate flood extents and water depths for urban and rural catchments.
- Conducted technical reviews of **hydraulic modelling studies** performed by junior team members, ensuring adherence to industry best practices and project specifications.
- Mentored and coached **less experienced team members** on hydraulic modelling software and project delivery methodologies, fostering professional growth.
- Generated detailed **technical reports** and visualisations using **QGIS** to communicate complex modelling results to clients and regulatory bodies.

SKILLS

Hydraulic Modelling – TUFLOW, TUFLOW FV, Flood Modeller, ESTRY, Flood Risk Management

GIS Software – QGIS, ArcGIS

Project Management – Client Management, Business Development, Proposal Writing, Stakeholder Engagement

Engineering Domain – Civil Engineering, Hydrology, Environmental Engineering, Water Quality Modelling, Sediment Transport

Reporting & Analysis – Technical Reporting, Problem-Solving, Data Analysis, Green Energy Infrastructure

PROJECTS

Urban Flood Risk Assessment (HCL Technologies)

- Developed detailed **hydraulic models** for a major urban area using **TUFLOW** to assess flood risk under various climate change scenarios, identifying critical infrastructure vulnerabilities.
- Integrated hydrological data with **GIS datasets** in **QGIS** to define model boundaries, rainfall inputs, and output analysis zones, supporting a **flood planning guidelines** compliance report.
- Generated comprehensive **technical reports** and maps detailing flood extents, depths, and velocities, which were presented to local authority stakeholders.

Riverine Floodplain Modelling (HCL Technologies)

- Built and calibrated a **Flood Modeller** model for a significant river system, simulating flow dynamics and floodplain inundation over a 50 km reach.
- Performed sensitivity analysis on key model parameters, including roughness coefficients and flow inputs, to understand model behaviour and uncertainty.
- Provided technical input for a client proposal focused on assessing the impact of proposed infrastructure on the riverine floodplain.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)